

My research interests include designing scientific **visualization** techniques for novel domain-specific applications. Beyond the traditional desktop setup, I am also interested in **immersive** analytics of scientific data, which uses **Augmented and Virtual Reality** to support visual analysis and decision-making.

Education

- 2015 – 2023 **Ph.D. in Computer Science**, Stony Brook University, NY, USA.
Dissertation: Immersive and Augmented 3D Scientific Visualization Systems
Advisor: Arie E. Kaufman
- 2010 – 2014 **B.S. in Computer Science**, School of Science and Engineering, LUMS, Lahore, Pakistan

Academic Appointments

- Aug 2025 – **Assistant Professor**, *Department of Computer Science*, University of Illinois, Chicago, IL, USA.
Now Electronic Visualization Lab
- Sept 2023 – **Principal Research Scientist**, Stony Brook University, NY, USA.
July 2025 Center for Visual Computing
- Aug 2016– **Graduate Research Assistant**, *Center for Visual Computing*, Stony Brook University, NY, USA.
Sept 2023 *Advisor*: Arie E. Kaufman
- 2014 – 2015 **Assistant Research Scientist**, *Technology for People Initiative*, LUMS, Pakistan.
Project: Digital Preservation of Heritage Sites
- 2012 – 2014 **Student Research Assistant**, *Computer Vision Lab*, LUMS, Pakistan.
Advisors: Sohaib A. Khan and Murtaza Taj

Academic Service

Conference Organization

IEEE VR, Organization Committee, Publication Chairs (2026, 2027)
IEEE VIS, Program Committee – 2024, 2025, 2026
ACM DIS, Program Committee, Artifacts and Systems, Associate Chair – 2026
IEEE VIS, Organization Committee, Supporter's Logistics – 2024
EuroVis, Program Committee – 2025
ACM IUI, Program Committee – 2025
IEEE VIS, Student Volunteer – 2018, 2022, 2023

Reviewer

- Journal Transactions on Visualization and Computer Graphics, IEEE
Transactions on Emerging Topics in Computing, IEEE
Scientific Reports, Nature Publishing
Information Processing and Management, Elsevier
International Journal of Human-Computer Interaction
- Conference IEEE VIS
IEEE Virtual Reality and 3D User Interface
SIGGRAPH Posters Program
IEEE Pacific Visualization Symposium
IEEE International Symposium on Mixed and Augmented Reality
ACM CHI Conference on Human Factors in Computing Systems

Eurographics
China Visualization and Visual Analytics Conference
Other IEEE Computer Graphics and Applications

Teaching Experience

- Fall 2025 – **Instructor**, CS 428, *Virtual Reality, Augmented Reality, Mixed Reality*.
Now CS Undergraduate and Graduate course, University of Illinois Chicago
- Spring 2026 **Instructor**, CS 427, *Creative Coding*.
CS Undergraduate and Graduate course, and CS+DES Undergraduate course, University of Illinois Chicago
- Spring 2024 **Instructor**, CSE 366, *Introduction to Virtual Reality*.
Undergraduate course of 35 students, Stony Brook University
- Spring 2024 **Instructor**, EMP 532, *Big Data Systems for Technology Management*.
Graduate course of 11 MS and professional students, Stony Brook University
- Spring 2019 – **Teaching Fellow**, CSE 566, *Virtual Reality*.
2023 Graduate course of 25 students, Stony Brook University
- Fall 2015 **Teaching Assistant**, CSE 114, *Object Oriented Programming*.
Undergraduate course of 100 students, Stony Brook University
- Spring 2014 **Teaching Assistant**, CS 200, *Data Structures*.
Undergraduate course of 100 students, LUMS

Grants

Grants Awarded at University of Illinois Chicago

- 2026 **NSF Future CoRe: HCC: Rethinking Visualization Evaluation through Image-Based Models of Human Vision**. PI: Khairi Reda, Co-PI: Saeed Boorboor. Award amount: \$500,000. Project period: 10/01/2026–09/30/2029.

Grants Awarded at Stony Brook University

- 2024 **Targeted Research Opportunity Program: VDR-Fusion Award: AI-Enhanced Multimodal Imaging for the Prevention of Retained Surgical Items**. PIs Georgios Georgakis, Todd Griffin, Saeed Boorboor, Manoj Mahajan, and Rong Zhao. Award amount: \$40,000 for 2 years.

Publications

Peer-reviewed Journal Papers

- 2026 [J11] Hossein Fathollahian, Siyuan Zhao, Nafiul Nipu, Marziye Salahshour, Lei Duan, Wei Tang, **Saeed Boorboor**, and G Elisabeta Marai. ConGAT: Context-aware graph attention network for 3d region of interest discovery in multiplexed microscopy images. *Computers & Graphics*, page 104728, 2026
- 2025 [J10] Yoonsang Kim, Zainab Aamir, Mithilesh Singh, **Saeed Boorboor**, Klaus Mueller, and Arie E Kaufman. Explainable XR: Understanding user behaviors of XR environments using LLM-assisted analytics framework. *IEEE Transactions on Visualization and Computer Graphics*, 2025
- 2023 [J9] **Saeed, Boorboor**, Matthew S Castellana, Yoonsang Kim, Chen Zhu-Tian, Johanna Beyer, Hanspeter Pfister, and Arie E Kaufman. VoxAR: Adaptive visualization of volume rendered objects in optical see-through augmented reality. *IEEE Transactions on Visualization and Computer Graphics*, 30(10):6801–6812, 2023
- 2023 [J8] **Saeed, Boorboor**, Yoonsang Kim, Ping Hu, Josef M Moses, Brian A Colle, and Arie E Kaufman. Submerge: Visualizing storm surge flooding simulations in immersive display ecologies. *IEEE Transactions on Visualization and Computer Graphics*, 30(9):6365–6377, 2023
- 2023 [J7] Ping Hu, **Saeed Boorboor**, Joseph Marino, and Arie E Kaufman. Geometry-Aware Planar Embedding of Treelike Structures. *IEEE Transactions on Visualization and Computer Graphics*, 29(7):3182–3194, 2023

- 2023 [J6] **Saeed Boorboor**, Shawn Mathew, Mala Ananth, David Talmage, Lorna W. Role, and Arie E. Kaufman. NeuRegenerate: A Framework for Visualizing Neurodegeneration. *IEEE Transactions on Visualization and Computer Graphics*, 29(3):1625–1637, 2023
- 2023 [J5] Brian A Colle, Julia R Hathaway, Elizabeth J Bojsza, Josef M Moses, Shadya J Sanders, Katherine E Rowan, Abigail L Hills, Elizabeth C Duesterhoeft, **Saeed, Boorboor**, Arie E Kaufman, and Susan E Brennan. Risk perception and preparation for storm surge flooding: A virtual workshop with visualization and stakeholder interaction. *Bulletin of the American Meteorological Society*, 2023
- 2022 [J4] Johanna Beyer, Jakob Troidl, **Saeed Boorboor**, Markus Hadwiger, Arie Kaufman, and Hanspeter Pfister. A Survey of Visualization and Analysis in High-Resolution Connectomics. *Computer Graphics Forum*, 41(3):573–607, 2022
- 2021 [J3] Parmida Ghahremani, **Saeed Boorboor**, Pooya Mirhosseini, Chetan Gudisagar, Mala Ananth, David Talmage, Lorna W Role, and Arie E Kaufman. Neuroconstruct: 3D reconstruction and visualization of neurites in optical microscopy brain images. *IEEE Transactions on Visualization and Computer Graphics*, 28(12):4951–4965, 2021
- 2019 [J2] Ji Hwan Park, Saad Nadeem, **Saeed Boorboor**, Joseph Marino, and Arie Kaufman. CMed: Crowd analytics for medical imaging data. *IEEE Transactions on Visualization and Computer Graphics*, 27(6):2869–2880, 2019
- 2018 [J1] **Saeed Boorboor**, Shreeraj Jadhav, Mala Ananth, David Talmage, Lorna Role, and Arie Kaufman. Visualization of neuronal structures in wide-field microscopy brain images. *IEEE Transactions on Visualization and Computer Graphics*, 25(1):1018–1028, 2018

Peer-reviewed Conference Papers

- 2026 [C9] Zainab Aamir, Ahamed Shoaib, **Saeed Boorboor**, and Arie Kaufman. FlexiCAVE: A Dynamically Configurable High-Resolution Display Facility. *PACMHCI Interactive Surfaces and Spaces (ISS)*. 2026
- 2025 [C8] Matthew S Castellana, Chahat Kalsi, Yoonsang Kim, **Saeed Boorboor**, and Arie E Kaufman. AuxiScope: Handheld augmented reality tablet as an auxiliary display for large-scale display systems. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 1660–1670, 2025
- 2025 [C7] **Saeed Boorboor**, Doris Gutiérrez-Rosales, Ahamed Shoaib, Chahat Kalsi, Yue Wang, Yuyang Cao, Xianfeng Gu, and Arie E Kaufman. Silo: Half-gigapixel cylindrical stereoscopic immersive display. In *IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pages 483–493, 2025
- 2023 [C6] Josef Moses, Brian Anthony Colle, Julia Roberts Hathaway, Elizabeth Bojsza, Shadya Sanders, Katherine E Rowan, Abigail Hills, Elizabeth C Duesterhoeft, **Saeed, Boorboor**, Arie E Kaufman, et al. A virtual workshop involving college students to explore the relative role of visualization and stakeholder interaction on risk perception. In *Conference on Weather Analysis and Forecasting, Numerical Weather Prediction, and Mesoscale Processes*. AMS, 2023
- 2022 [C5] Ping Hu, **Saeed Boorboor**, Shreeraj Jadhav, Joseph Marino, Seyedkoosha Mirhosseini, and Arie E Kaufman. Spatial Perception in Immersive Visualization: A Study and Findings. In *International Symposium on Mixed and Augmented Reality Adjunct*, pages 369–372. IEEE, 2022
- 2021 [C4] Yoonsang Kim, **Saeed Boorboor**, Amir Rahmati, and A Kaufman. Design of privacy preservation system in augmented reality. In *IEEE Symposium on Visualization for Cyber Security*, 2021
- 2018 [C3] Mala Ananth, **Saeed Boorboor**, Shreeraj Jadhav, Arie Kaufman, Mark Slifstein, Nikhil Palekar, Christine DeLorenzo, Ramin Parsey, David Talmage, and Lorna Role. Visualizing the cholinergic system in health & disease. In *Neuropsychopharmacology*, volume 43, pages S263–S264. Nature Publishing Group, 2018
- 2018 [C2] **Saeed Boorboor**, Saad Nadeem, Ji Hwan Park, Kevin Baker, and Arie Kaufman. Crowdsourcing lung nodules detection and annotation. In *Medical Imaging: Imaging Informatics for Healthcare, Research, and Applications*, volume 10579, pages 342–348. SPIE, 2018
- 2015 [C1] Ahsan Abdullah, Reema Bajwa, Syed Rizwan Gilani, Zuha Agha, **Saeed Boorboor**, Murtaza Taj, and Sohaib Ahmed Khan. 3D Architectural Modeling: Efficient RANSAC for n-gonal Primitive Fitting. In *Eurographics*, pages 5–8, 2015

Peer-reviewed Workshops, Posters, and Abstracts

- 2026 [W3] Chahat Kalsi, Hossein Fathollahian, Luca Reichmann, Sophia Gaupp, Yuancheng Shen, Aleksandra Shabanova, Ada Junquera, Angéla Szabó, Alfredo Rodríguez, Tuulia Vallius, Meri Rogava, G. Elisabeta Marai, Michael Krone, **Saeed Boorboor**, Robert Krüger. Scalable 3D Visualization and Spatial Exploration of Multiplexed Cancer Tissue Images for the Web. *3rd Conference of the European Society for Spatial Biology (ESSB) Poster Program, Oct 2026.* 🏆
- 2026 [W2] Ava Nederlander, Matthew Castellana, **Saeed Boorboor**, Arie Kaufman, and Georgios Georgakis. Abdominal Cavity Reconstruction from Laparoscopic PIPAC Video: Toward Quantitative and Qualitative Assessment of Peritoneal Surface Disease. *Society of Surgical Oncology (SSO), 20th Advanced Cancer Therapies, 2025. Selected for Oral Presentation.*
- 2025 [W1] Chahat Kalsi, Yuancheng Shen, Sophia Gaupp, and Luca Reichmann, Meri Rogava, Michael Krone, **Saeed Boorboor**, and Robert Krüger. BioSET - Biomarker-based Spatial co-Expression analysis in Tumor environments. *In IEEE BioMedVis Challenge 2025, IEEE VIS, 2025.* 🏆

Patents and Licenses

- P1 System, Method, and Computer-Accessible Medium for Processing Brain Images and Extracting Neuronal Structures. A Kaufman and **S Boorboor**. US Patent App. 16/994,885.
- P2 Method and System for Color Rendering in Augmented and Mixed Reality. A Kaufman and **S Boorboor**. Copyrighted software to facilitate the visualization of virtual objects from their environment in augmented and mixed reality. Technology ID: 050-9419.

Mentoring Experience

Doctoral Students

- 2026 – Now **Rehab Uddin Shwon**, *University of Illinois Chicago.*
- 2026 – Now **Rob Kondratowicz**, *University of Illinois Chicago.*
- 2026 – Now **Hossein Fathollahian**, (*co-advisor*), *University of Illinois Chicago, [J11, W1, W3].*
- 2023 – 2025 **Zainab Aamir**, *Stony Brook University, [C9, J10].*
- 2021 – 2025 **Yoonsang Kim**, *Stony Brook University, [J8, J9, J10, C1].*
- 2021 – 2025 **Matthew Castellana**, *Stony Brook University, [J8, J9, C8].*

Masters Students

- 2025 – Now **Riccardo Bonfanti**, *University of Illinois Chicago.*
- 2024 – 2025 **Chahat Kalsi**, *Stony Brook University, [C7, C8, W1].*
- 2024 – 2025 **Ahamed Shoaib**, *Stony Brook University, [C7].*

Awards and Honors

- 2024 Recognition by the National Academy of Inventors - Chapter for Inventions at Stony Brook University.
- 2019, 2021 IACS Junior Researcher Award. Awarded for strong contributions in computational research. (\$37,000 + \$5000 travel grant.)
- 2016 Stony Brook University CS Department Chair Fellowship: Awarded the special chair fellowship for the top 10 Ph.D. applications. (\$10,000)

Selected Talks and Presentations

- 2025 Invited Lecture on Situated Visualization, *New York University, USA.*
- 2024 Visual Analysis of Volumetric Videos, *Harvard University, USA.*
- 2022 Visualization in Immersive Display Ecologies, *Center for Visual and Decision Informatics.*
- 2018 Visualization of Neuronal Structures in Optical Microscopy, *Harvard University, USA.*
- 2018 Visualization for Neuroscience and Connectomics. *LUMS, Pakistan.*